

results in today's lecture." "There is money in it—let me take out [a] patent for you. You do not know what money you are throwing away," said Page. Of course, "I will only take half share in the profit—I will finance it," he said. This multimillionaire was pleading like a beggar.

The scientist refused the offer and delivered his lecture at the Royal Society.

This gentleman was Jagdish Chandra Bose!

Pinging

'Ping' was written by Mike Muuss in December 1983.

From his home page, "I'm the architect of BRL-CAD, a substantial third-generation CSG solid modeling system, available free of charge, which you probably have never heard of, and the author of ping, tcp and assorted other network goodies. Ping is a little thousand-line hack that I wrote, which practically everyone seems to know about. It's included in every copy of UNIX and Windows, putting it into nearly every computer on the planet."

Sadly, Mike was killed in an automobile accident on November 20, 2000. His home page is still available, a testament to his intellect and indomitable spirit.

Home of World Wide Web

His parents taught him to use mathematics everywhere, even at the dining table. Out of habit, he played with imaginary numbers even while eating. As a hobby, he built a computer with TTL gates and M6800 processor and an old television while still studying at the Queen's College at the Oxford University, England.

To solve the problem of sharing information at Large Hadron Collider at CERN, he created the World Wide Web. He did this single-handedly, unlike most other scientists. And he did not patent it. The man behind this information explosion is Tim Berners-Lee.

Eric Schmidt, CEO, Novell, commented in the New York Times, "If this were a traditional science, Berners-Lee would win a Nobel Prize. What he has

done is that significant."

Commenting on the home page in the Internet, Tim Berner said, "They may call it a home page, but it's more like the gnome in somebody's front yard than the home itself."

Double Nobel

John Bardeen, Walter H. Brattain and William Shockley were awarded the Nobel Prize in Physics in 1956, for 'investigations on semiconductors and the discovery of the transistor effect.'

The Nobel ceremony took place in Sweden on the evening of Monday, December 10. John Bardeen did not want to disturb the studies of his two sons at Harvard. So he brought his third (only other) son to the Nobel ceremony. King Gustav chided Bardeen for leaving his family behind on such an important occasion. With profound confidence, he assured the King that the next time he would bring all his children, as if he would bring them for the next-time dinner.

In 1972, Bardeen was awarded the Nobel again, along with L.N. Cooper and J.R. Schrieffer, for the successful explanation of superconductivity. And, he did bring all his children to that Nobel ceremony.

John Bardeen is the only person in history to have received two Nobel Prizes in physics.

Dr X

There was this popular TV quiz show in 1957, 'I've Got a Secret', wherein audience were challenged to recognise a celebrity known as 'Dr X' with a few clues. One of the celebrity panelists, Bill Cullen, asked 'Dr X' if he had invented some kind of machine that might be painful when used. The mysterious doctor replied, "Yes, sometimes it's most painful."

Later the identity of 'Dr X' was revealed as Philo Farnsworth, the father of electronic television. Then the programme anchor confessed, "We'd all be out of work if it weren't for you."

Hearty mistake

Wilson Greatbatch was building an

oscillator to record heart sounds in the late 1950s. On powering up the circuit, he observed that it was giving a steady electrical pulse of 1.8 millisecond followed by a 1-second interval. A mistake somewhere! Recalling the incident, Greatbatch said, "The oscillator required a 10,000-ohm resistor at the transistor base. I reached into my resistor box for one, but I misread the colour coding and got a 1-megaohm resistor by mistake." He was about to desolder it, but he waited for a moment. These pulses and the device could regulate the heartbeat. After two years of refinement and re-refinement, he built the first successful implantable pacemaker.

A deeply religious man, he believed "It was no accident. The Lord was working through me."

Until then, these devices were the size of a television and as complicated those days. A Canadian, John Hopps invented the first cardiac pacemaker.

Maser revelation

On the early morning of April 26, 1951, Charles Townes, a microwave physicist, was walking his way to a conference in Washington, DC, with an unsolved problem in his mind. For a moment, he felt like sitting on the Franklin park bench, where ideas were pouring down on him. He pulled an envelope out of his jacket—the only piece of paper available with him then—and started jotting down the calculations. The result was the MASER (microwave amplification by stimulated emission), which amplifies microwaves to produce an intense beam and the development of LASER (light amplification by stimulated emission). Writing parallels between religion and science, later on, the physicist Charles Townes wrote, "There is a tremendous emotional experience (in scientific discovery) which I think is similar to what some people would normally describe as a religious experience, a revelation." ●

The author has written six science books published by Pustak Mahal, New Delhi, and an engineering book by Industrial Press, New York. He has contributed several articles to EFY in the past as well